

Can An Investor Profit from Return Predictability in Real Time?*

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Abstract

Despite the extensive empirical evidence on return predictability, the question whether investors can profit from return predictability in real time remains unsettled. One problem is that the predictive model is likely misspecified, which has received little attention. By taking a more careful approach that guards against model mis-specification, we study the economic significance of return predictability by evaluating extensively the out-of-sample portfolio performance of various predictive models and variations of predictive variables. We find that investors can benefit from predicting the sign, but not the magnitude, of the market expected returns only in periods when the market performs poorly. We also find that there is no significant performance difference among the predictive models, although the best statistical model often performs the worst and a GARCH feature seems important.

Introduction

Ever since Fama and Schwert (1977) and Keim and Stambaugh (1986), empirical literature has extensively documented that asset returns can be predicted to a certain extent by many economic variables.¹ More recent studies examine the economic implications of return predictability on investors' asset allocation decisions and find that investors behave differently when returns are predictable.² However, the question whether investors can profit from return predictability is still largely unsettled. Studies such as Kandel and Stambaugh (1996), Balduzzi and Lynch (1999), Lynch (2001), Gomes (2002), and Campbell, Chan, and Viceira (2002) evaluate the economic benefits via *ex ante* calibration and often find significant positive results. By contrast, studies that evaluate the *ex post* performance of return predictability, which is perhaps more appropriate to address the question than the *ex ante* calibration, find mixed results. Early studies such as Breen, Glosten, and Jagannathan (1989) and Pesaran and Timmermann (1995) find that return predictability yields significant economic gains out of sample. However, more recent studies such as Cooper, Gutierrez, and Marcum (2001) and Cooper and Gulen (2001) fail to find any economic significance. Similarly, studies such as Jacobsen (1999) and Marquering and Verbeek (2001) find that in a mean-variance framework the economic gains of exploiting return predictability are significant, whereas Handa and Tiwari (2004) find that the economic significance of return predictability is questionable. Furthermore, return predictability is also challenged on the statistical ground. It is well-known that the statistical significance of return predictability is weak and may be subject to estimation bias.³ In addition, focusing on the out-of-sample forecasting, studies such as Bossaerts and Hillion (1999), Neely and Weller (2000), and Goyal and Welch (2003), fail to find any significant statistical evidence of return predictability.

One potential problem is that the predictive model is mis-specified. As the empirical literature has relied on ad hoc search for powerful predictive variables, much of the attention in the literature is focused on the choice of predictive variables (see, e.g. Pesaran and Timmermann, 1995; Bossaerts and Hillion, 1999; Avramov, 2002; Cremers, 2002; Roskelley, 2004). However, model specification of the underlying data-generating process has received little attention. In most previous studies, the predictive model is

¹See, e.g., Campbell (1987), French, Schwert, and Stambaugh (1987) Campbell and Shiller (1988a), Campbell and Shiller (1988b), Fama and French (1988), Fama and French (1989), Ferson (1989) Harvey (1989), Schwert (1989), Jegadeesh (1990), Harvey (1991), Ferson and Harvey (1991), Cochrane (1991), Hodrick (1992), Bekaert and Hodrick (1992), Lamont (1998), Lewellen (1999), Lettau and Ludvigson (2001), Shanken and Tamayo (2001), Santos and Veronesi (2001), Cremers (2002), Avramov (2002), and Goyal and Welch (2003).

²See, e.g., Barberis (2000), Campbell, Chan, and Viceira (2002), Balduzzi and Lynch (1999), Lynch (2001), Brennan, Schwartz, and Lagnado (1997), Guidolin and Timmermann (2002)

³See, e.g., Nelson and Kim (1993), Goetzmann and Jorion (1993), Kirby (1997), Stambaugh (1999), Ferson, Sarkissian, and Simin (1999), and Ang and Bekaert (2001).

specified as a first-order vector autoregression (VAR) or a further simplified predictive linear regression, which is subject to estimation bias discussed by Ferson, Sarkissian, and Simin (1999) and Stambaugh (1999). Nevertheless, this simple model is most likely mis-specified because returns and the predictive variables have very different dynamics (e.g., predictive variables are highly auto-correlated). In this study, we take a more careful approach that guards against model mis-specifications and evaluate the economic value of return predictability out of sample.

Specifically, we examine the out-of-sample portfolio performance of an array of predictive models ranging from the simplest VAR model to the nonlinear nonparametric model, whose specifications are chosen according to several statistical criteria. A serious limitation of the VAR model is the assumption of constant variance that is clearly inconsistent with the large empirical literature that documents time-varying volatility in asset returns, which also suggests another important source of predictability ignored by the VAR model, that is, the predictability of the conditional second moments of asset returns. To overcome this restriction, we extend the VAR model to incorporate GARCH effects, which we refer to as the VAR-GARCH model. To our best knowledge, our analysis seems to be the first application of the VAR-GARCH model to the area of return predictability even though it has been applied in other areas such as dynamics of interest rates and exchange rates where the focus is on the GARCH effect.⁴ This extension allows us to consider the combined effect of predicting both return and volatility, and to obtain more accurate estimation of the conditional distributions of returns, which *a priori* may result in better portfolio decisions.

Nevertheless, both the VAR and VAR-GARCH models are restricted by two assumptions. The first is that both models assume the (conditional) distributions of returns are normally distributed.⁵ As a result, only mean (VAR) and variance (VAR-GARCH) need to be modeled. The second assumption is that asset returns are linear functions of predictive variables. However, it is likely that the true relations between asset returns and predictive variables are highly nonlinear, and hence the estimated conditional distributions of returns are likely biased. To overcome these limitations, we consider the seminonparametric (SNP) models proposed by Gallant and Tauchen (1989), which rely on Hermite polynomial expansions to approximate the underlying data-generating process, and thus is capable of capturing more features of the data. In addition, the SNP model nests the VAR and VAR-GARCH models as special cases, thus facilitating model comparison and selection. As in Bossaerts and Hillion (1999), to further guard against possible model mis-specification,

⁴Recently Tang (2003) uses the VAR-GARCH model as the true data-generating process to evaluate the performance of Bayesian model averaging methodology introduced to the literature by Avramov (2002) and Cremers (2002).

⁵To be fair, it should be noted that GARCH models can incorporate non-normal error terms.

we use statistical model selection criteria to choose the best specification.

We conduct portfolio analysis for two different types of risk averse investors, whose preferences are represented by the power utility, who allocate funds to the market portfolio (S&P 500 index) and the 1-month Treasury bill to maximize their expected utilities. The first investor believes that asset returns are predictable and forms her optimal portfolios based on the predictive models considered in this study. As a benchmark, the second investor believes that asset returns can only be described by random walks and chooses her optimal portfolio weights according to a simple normal I.I.D model. The portfolio analysis is conducted both in sample and out of sample, and we consider the cases with or without imposing the no-short-sale constraint. The portfolio performance is measured *ex post* by various performance measures including the Sharpe ratio, a measure proposed by Graham and Harvey (1997), and the certainty equivalent rate of return (CER).⁶

We consider a number of variations of predictive variables, although incorporating model uncertainty in the aspect of choosing predictive variables is beyond the scope of this study. Among the numerous empirically identified predictive variables, the most popular one is the dividend yield as it is backed by theory to some extent. Many studies including Kandel and Stambaugh (1996), Barberis (2000), and Balduzzi and Lynch (1999), all use the dividend yield as the predictor. Other variables we choose are 3-month T-bill yield, term spread (yield difference between a 10-year Treasury Bond and a 1-year Treasury Bill), and default spread (yield difference between Moody's AAA-rated and BAA-rated corporate bonds), all of which are shown to be powerful predictors of stock returns.

We find several interesting results. First, we find strong in-sample evidence that portfolios based on the predictive models significantly outperform the benchmark portfolios based on a fixed-weight strategy and passive buy-and-hold strategy. This is true for all variations of predictive variables considered except the dividend yield, in which case only the VAR model outperforms the benchmarks. However, evidence of superior out-of-sample performance is very limited. Indeed, no portfolios based on any predictive models outperform the benchmark portfolios when no portfolio constraint is imposed regardless of the predictive variables present. Superior performance is possible only when the no-short-sale constraint is imposed and when the T-bill yield is present. Similar results are obtained when we estimate the predictive models recursively to capture possible structure and parameter changes. Incorporating appropriate transaction costs in the investor's objective functions does not change the results significantly. The out-of-sample results

⁶This is in contrast to studies such as Kandel and Stambaugh (1996) and Pástor and Stambaugh (2000), using *ex ante* CERs to measure *ex ante* portfolio performance.

suggest that predictive variables may correctly predict the sign of the market expected returns but not the magnitude. This is further supported by even stronger performance of the switching portfolios that switch between the market portfolio and 1-month T-bill depending on whether or not the forecasted expected excess returns are positive.

Second, even when the portfolio weights are constrained, only T-bill yield outperforms the benchmarks,⁷ whereas the default spread, term spread, and in particular dividend yield, do not seem to possess any predictive power. On the other hand, the default spread and term spread seem to complement T-bill yield in that combining each with T-bill yield produces stronger performance. It is interesting to note that dividend yield has no predictive power either in sample or out of sample, which suggests that future research should avoid using this variable.⁸

Third, the performance differences among the predictive models are not significant, although the VAR-GARCH model often performs the best, which suggests that it may be necessary to incorporate conditional heteroscedasticity in the predictive model. In a related study, Carlson, Chapman, Kaniel, and Yan (2004) examine the utility cost of ignoring volatility dynamics through calibration analysis and find that volatility related specification error is economically significant. On the other hand, an overfitted nonlinear and nonparametric SNP model also performs well and sometimes the best. The best model according to the statistical criteria, however, performs the worst most of the time. This result underscores the importance of choosing the appropriate model selection criterion. For portfolio analysis, one should use a relevant financial criterion not a statistical criterion. It also suggests that in-sample inference may not be relevant to out-of-sample performance.

Fourth, the relative performance of the predictive variables and predictive models are unstable over time. In particular, the predictive models tend to outperform the benchmarks when the market performs poorly (volatility is high and Sharpe ratio is low) and underperform the benchmarks when the market performs well (volatility is low and Sharpe ratio is high).

Finally, our results are very robust. In light of the criticism brought up by Cooper and Gulen (2001), we conduct extensive robustness analysis and obtain similar results when we vary the number of years in the recursive estimation, the investment horizon, the degree of risk aversion, and the utility function.

⁷Ang and Bekaert (2001) also find that the short rate is the only one having the statistically significant coefficient in a predictive regression with dividend yield and earning yield.

⁸Both Ang and Bekaert (2001) and Goyal and Welch (2003) also find that dividend yield has no predictability in statistical tests.

Our findings may explain the mixed results obtained by previous studies. For example Breen, Glosten, and Jagannathan (1989), Pesaran and Timmermann (1995), and Abhyankar and Davies (2002) find out-of-sample economic significance using the switching rule. Marquering and Verbeek (2001) find significant gains from predicting market returns when short-selling is prohibited, whereas Handa and Tiwari (2004) find that no significantly superior performance exists when (limited) short-selling is allowed. On the other hand, none of the studies examine the portfolio performance simultaneously when short-selling is restricted and when short-selling is allowed. By comparing the performance of the two cases, we are able to show that predictive variables such as T-bill yield can only predict the direction of market changes but not the magnitude of the changes, which is a very interesting and new result. In addition, Pesaran and Timmermann (1995) find better predictive performance in the volatile period of 1970's, and Handa and Tiwari (2004) find poor predictive performance in the most recent period after 1989 whereas favorable results during the periods of 1959–1973 and 1974–1988. Abhyankar and Davies (2002) find that the predictive ability of T-bill yield is low prior to the 1951 Treasury Accord Act period, high during the period of 1950-1975, and has disappeared in the last two decades. Our finding that the predictive performance is related to the market condition is largely consistent with these findings.

The remainder of this article is organized as follows. Section 1 discusses investor's optimal portfolio choice problem. Section 2 discusses the group of predictive models that incorporate return predictability. Section 3 describes the predictive variables and discusses the estimation results of the predictive models. Section 4 conducts in-sample as well as out-of-sample portfolio analysis to examine the performance of the various predictive models. Section 5 concludes.

1 Investor's Portfolio Choice Problem

Assume a risk-averse investor has a preference over wealth represented by a utility function $u(W)$, where W is her wealth. The investor chooses her asset allocation policy between a risky asset (the market portfolio) and a riskless asset (e.g. 30-day Treasury Bill), to maximize her expected utility given the conditional distributions of asset returns.

Specifically, the investor solves the following one-period optimization problem at time t :

$$\max_{\omega_t} \mathbf{E}[u(W_{t+1})|\mathcal{F}_t] = \max_{\omega_t} \int u(W_{t+1})p(r_{t+1}|\mathcal{F}_t)dr_{t+1}, \quad (1)$$

s.t.

$$W_{t+1} = W_t[\omega_t e^{r_{t+1} + r_{ft}} + (1 - \omega_t) e^{r_{ft}}],$$

where r_{t+1} and ω_t are the future excess return and portfolio weight on the market portfolio, respectively, and r_{ft} is the return on the riskless asset.

The integration in eq. (1) is evaluated numerically by Monte Carlo simulation. The optimization problem can be written as

$$\max_{\omega_t} \frac{1}{N} \sum_{i=1}^N u(W_t[\omega_t e^{r_{t+1|t}^{(i)} + r_{ft}} + (1 - \omega_t) e^{r_{ft}}]). \quad (2)$$

If we assume that the investor's preference over wealth is determined by the constant relative risk averse power utility, then the optimization problem is

$$\max_{\omega_t} \frac{1}{N} \sum_{i=1}^N \frac{W_t^{1-\gamma} [\omega_t e^{r_{t+1|t}^{(i)} + r_{ft}} + (1 - \omega_t) e^{r_{ft}}]^{(1-\gamma)}}{1 - \gamma},$$

where γ is the investor's relative risk aversion coefficient, $r_{t+1|t}^{(i)}$ are the one-step ahead predictive sample draws generated from the predictive models, which will be discussed in next section, and N is the number of sample draws. The optimization is solved numerically by Brent method with analytic derivatives.

In addition to the one-period problem, we also examine portfolio performance at longer investment horizons, for example, 3 months or 6 months. In general, if the investor has an investment horizon of $\hat{T}$ periods, then the forecast of the one-period excess return $r_{t+1|t}$ is replaced by the forecast of the cumulative excess return over the T periods, $R_{t+\hat{T}|t}$,

$$R_{t+\hat{T}|t} = r_{t+1|t} + r_{t+2|t} + \cdots + r_{t+\hat{T}|t}, \quad (3)$$

and the investor's problem is to solve

$$\max_{\omega_t} \frac{1}{N} \sum_{i=1}^N \frac{W_t^{1-\gamma} [\omega_t e^{R_{t+\hat{T}|t}^{(i)} + r_{f\hat{T}}} + (1 - \omega_t) e^{r_{f\hat{T}}}]^{(1-\gamma)}}{1 - \gamma}. \quad (4)$$

In the presence of transaction costs, the investor will choose the optimal portfolio weights taking into consideration the costs associated with rebalancing the weights. Assume the proportional transaction cost is τ for the market portfolio, and there is no transaction cost in trading the riskless asset. The investor's

wealth is given by

$$W_{t+1} = W_t(1 - f_t)[\omega_t e^{r_{t+1} + r_{ft}} + (1 - \omega_t)e^{r_{ft}}], \quad (5)$$

where the transaction cost at time t , f_t , is given by

$$f_t = \tau |\omega_t - \hat{\omega}_t|, \quad (6)$$

and $\hat{\omega}_t$ is the inherited portfolio weight from the previous period,

$$\hat{\omega}_t = \frac{\omega_{t-1} e^{r_t + r_{ft-1}}}{\omega_{t-1} e^{r_t + r_{ft-1}} + (1 - \omega_{t-1}) e^{r_{ft-1}}} \quad (7)$$

2 Predictive Models

The vector autoregression model (VAR) has been extensively used in the literature to model return predictability of the market portfolio. It captures the basic notion that the mean of return is a (linear) function of the predictive variables. The general specification of the VAR model is given by,

$$\mathbf{y}_t = \Phi_0 + \sum_{i=1}^{L_\mu} \Phi_i \mathbf{y}_{t-i} + \epsilon_t, \quad (8)$$

where $\mathbf{y}_t$ is the state vector including the excess returns on the market portfolio and predictors at time t , and ϵ_t is a vector of normally distributed disturbances with a zero vector of means and variance-covariance matrix Σ . L_μ denotes the order of autoregression and is never higher than one in empirical studies. Many studies often use a further simplified predictive linear regression, which only considers the return equation in the VAR model, but this regression model is subject to estimation bias discussed by Ferson, Sarkissian, and Simin (1999) and Stambaugh (1999).

The VAR model assumes the disturbances ϵ_t are independently identically distributed. Stock returns, however, exhibit prominent conditional heteroscedasticity (see, e.g., French, Schwert, and Stambaugh, 1987; Engle, 1982). Therefore, a natural extension of the VAR model to deal with conditional heteroscedasticity is to incorporate GARCH features. The extended VAR-GARCH model captures predictability in both the first and the second moments of stock returns. However, both VAR and VAR-GARCH models assume conditional normality of return distributions and a linear relation between returns and predictive variables. To further relax these assumptions, we consider the seminonparametric (SNP) model proposed by Gallant and Tauchen

(1989). The SNP model relies on the Hermite polynomial expansions to approximate the conditional density of the underlying data-generating process. Because of polynomial expansions the conditional distribution is no longer normal, and the moments are nonlinear functions of the predictors. Another relevant advantage is that the SNP nonlinear model nests both the VAR and VAR-GARCH models, which makes it possible to compare different models statistically in the classical paradigm. To facilitate estimation and model comparison, all models including the VAR and VAR-GARCH models are estimated using the procedure proposed by Gallant and Tauchen (1997).

The SNP model is specified as follows. Let $f(\mathbf{y}|\mathbf{x}, \theta)$ denote the conditional density of the state vector $\mathbf{y}$ conditioned on the lagged values of $\mathbf{y}$, denoted by $\mathbf{x}$. Then

$$f(\mathbf{y}|\mathbf{x}, \theta) \propto [P(\mathbf{z})]^2 \phi(\mathbf{y}|\mu_x, \Sigma_x), \quad (9)$$

where

$$\begin{aligned} \mathbf{z} &= R_x^{-1}(\mathbf{y} - \mu_x), \quad \mu_x = b_0 + Bx, \quad \Sigma_x = R_x R_x', \\ Vec(R_x) &= \rho_0 + \sum_i^{L_r} \rho_i |\mathbf{y} - \mu_x| + \sum_j^{L_g} Diag(G_j) Vec(R_j), \end{aligned}$$

and $P(\mathbf{z})$ is the multivariate Hermite polynomials with degree K_z . The GARCH specification used in the SNP model is more akin to the one suggested by Nelson (1991). It is easy to see that when K_z is zero, the Hermite polynomial degenerates to a constant, and thus the SNP model degenerates to the VAR-GARCH model; when Σ_x is constant, the SNP model further degenerates to the Gaussian VAR model. Note that because of the rich parameterizations in multivariate GARCH, we restrict the GARCH to a diagonal specification. For financial data, it may be necessary to consider a more general model where the coefficients of the polynomial $P(\mathbf{z})$ are a polynomial of degree K_x in x because of the extraordinary heteroscedasticity. This model is fully nonlinear and nonparametric. Collectively, the parameters L_μ , L_g , L_r , K_z , and K_x uniquely identify the SNP model,⁹ and hence we use “ $(L_\mu, L_g, L_r, K_z, K_x)$ ” to denote the specification of a predictive model. For example, $(1, 0, 0, 0, 0)$ denotes VAR(1), while $(1, 1, 1, 0, 0)$ denotes VAR(1)-GARCH(1,1).

One advantage of our framework is that we systematically select the best model specification for each model (VAR, VAR-GARCH, and SNP models) using statistical criteria. Another advantage is that the

⁹There are two additional parameters, I_z and I_x , used to reduce the cross-interaction terms in the polynomials when y is multivariate. The highest order for cross-interaction terms is $K_z - I_z$ and $K_x - I_x$ respectively.

models are nested, which allows us to compare and select the best model specification across the different models. This systematical approach is far superior to the *ad hoc* assumption that the data follows certain process such as VAR(1). To this end, we use Schwarz’s Bayesian Information Criterion (BIC). Additional statistical criteria including Akaike’s Information Criterion (AIC) and Hannan and Quinn Criterion (HQ) are also considered. Because the BIC is conservative as it selects sparser parameterizations than the AIC, we also consider the fully nonlinear model, which is always rejected by the BIC due to rich parameterizations. In the sequel we denote the best SNP model OPT and the fully nonlinear and nonparametric model as NLNP. On this note, both Bossaerts and Hillion (1999) and Pesaran and Timmermann (1995) emphasize using statistic criteria to choose the best predictive models. Among others, the key difference between those two studies and this study is that they restrict to linear regression models, whereas we consider a more broad class of models including both linear and nonlinear ones.

3 Estimation of the Predictive Models

3.1 Data Description

In recent years, empirical literature has found more and more economic variables that have predictive power over stock and bond returns. These variables include term spread, dividend yield, Treasury-bill yield, default spread, consumption to wealth ratio (Lettau and Ludvigson, 2001), investment to capital ratio (Cochrane, 1991), dividend to earnings ratio (Lamont, 1998), debt to equity ratio (Schwert, 1989), and lagged returns, just to name a few. Among the predictive variables, dividend yield is the most popular one, and T-bill yield, term spread and default spread are found to be powerful predictors and are also widely used. In our empirical analysis, we use these four predictive variables as examples to illustrate our analysis, but the same analysis can be easily carried out with other predictive variables.

We use the S&P 500 composite index as the market portfolio. Monthly returns on the S&P 500 index and 30-day Treasury bill are obtained from the CRSP and converted to continuously compounded returns. Excess returns in percentage are used to fit various predictive models and converted to decimal returns for portfolio optimization. The dividend yield (DVYD) defined as the sum of the dividends paid on the S&P 500 index over the past 12 months divided by the current level of the index, the 3-month Treasury-bill yield (TBYD), the term spread (TRSD) defined as the difference in yield between the ten-year Treasury bond and one-year Treasury-bill, and the default spread (DFSD) defined as the difference in yield between Moody’s

AAA bonds and BAA bonds, are obtained from the DRI. Monthly data from January 1947 to December 1998 spanning 624 months are collected except for term spread, which is only available from April 1953, a total of 549 observations.

Panel A in Table 1 lists the mean, standard deviation, and other statistics about the market excess returns (EXRN), the returns on the 30-day Treasury bill (RFT1M), and the predictive variables. As expected, the excess returns exhibit negative skewness and excess kurtosis; Jarque-Bera statistics also indicate that the excess returns, riskfree rates, and the predictive variables are far from normally distributed. Also listed are the autocorrelation coefficients up to lag 12. The excess returns have very little autocorrelations, whereas the predictive variables are highly autocorrelated, with the 1st order autocorrelation coefficients as high as 0.989.

Panel A also reports statistics of the excess returns for two subperiods 1947:1–1978:12, and 1979:1–1998:12. In the out-of-sample analysis, we use the first period for parameter estimation and the second period for evaluating the out-of-sample performance.¹⁰ The excess returns display quite different dynamics in the two subperiods: The average excess return in the second subperiod is much higher than the first subperiod, and the excess returns are more skewed in the second subperiod than in the first subperiod.

Panel B in Table 1 lists the correlation matrices of the excess returns and the predictive variables in the whole sample period and the two subperiods.¹¹ It is worth noting that the correlations are not stable. For example, the correlation between the excess return and dividend yield is about -0.03 for the whole period but is positive (0.045) in the first subperiod and negative (-0.115) in the second subperiod. For the other two predictive variables, default spread and term spread, the correlations are much weakened in the second subperiod. On the other hand, the correlation between the excess return and T-bill yield is relatively stable and remains considerable. These differences in correlations are consistent with our portfolio performance results that T-bill yield is the strongest predictor, followed by term spread and default spread, and dividend yield does not have any predictive power at all. Interestingly, the correlations of the dividend yield with T-bill yield and default spread increase from negative in the first subperiod to positive in the second subperiod, whereas the correlation between default spread and term spread decreases to negative in the second subperiod. Other correlations also change over different periods.

¹⁰The two sub-periods are divided arbitrarily. The choice of 20 years as the out-of-sample testing period is to have enough samples to make statistically significant statements.

¹¹To include the term spread, the whole sample period starts from April, 1953. With other predictive variables, the whole sample period starts from January, 1947.

3.2 Specification Search and Model Estimation

The empirical literature on return predictability has been using the VAR(1) model or a further simplified predictive regression model as the data-generating process, which assumes linear relations between the market excess returns and predictive variables. However, little attention has been paid to the appropriateness of the assumed models. In this subsections, we examine different predictive models and try to identify the best model according to an array of statistical criteria.

We use the monthly time series of excess returns and predictive variables to search for the best specification for each of the following models: VAR, VAR-GARCH, and SNP. We consider various combinations of the four predictive variables. The best specification for each predictive model and each combination of predictive variables is selected according to statistic criteria such as Bayesian Information Criterion (BIC), Akaike's Information Criterion (AIC), and Hannan-Quinn Criterion (HQ).

Table 2 lists the best specifications for the whole sample period (Panel A: In-Sample Estimation) and the first subperiod (Panel B: Out-Of-Sample Estimation).¹² Several interesting results emerge. First, the VAR model, which is normally assumed in the literature, is clearly misspecified. Adding the GARCH specification substantially improves the fitness over the VAR model, as demonstrated by the much smaller values for all the criteria, whereas adding SNP specification only marginally improves the fitness. Second, The most popular VAR(1) model is not the best VAR model for most combinations of predictive variables but a VAR(2) is. Third, different combinations of predictive variables have different optimal specifications. For example, a GARCH(1,1) specification is optimal for default spread and term spread, but a GARCH(2,1) specification is the best for dividend yield and T-bill yield. This observation suggests that it may be inadequate in previous studies to choose different predictive variables while keeping the model fixed. Fourth, a polynomial of degree 4 is often the best choice for the SNP model, which is also the best specification overall because it yields the smallest BIC and often smallest HQ. As expected, the nonlinear and nonparametric model (NLNP) often has higher BIC value but the smallest AIC value. Fifth, two or more predictors provide better fit than any single one of them does. While the improvement is small compared to T-bill yield, it is considerable compared to term spread and default spread. For example, T-bill yield and term spread combined yield 0.16 for BIC with the OPT model, whereas T-bill alone yields 0.27 and term spread alone 1.29 for BIC. Finally, comparing Panel A and B, we find that in some cases the best specifications are different in the

¹²We do not show the combination of dividend yield with any other predictive variables because dividend yield, as we will show later, does not have any predictive power.

two periods. For example, the best GARCH specification for T-bill yield in the whole sample period is GARCH(2,1), whereas in the estimation period (1947:1–1978:12) it is GARCH(1,1). This suggests that the relations between the market returns and predictive variables may change over time. In addition, even when the specifications remain the same, the parameter estimates are different, suggesting that the parameters may change over time.

4 Portfolio Performance of the Predictive Models

Having determined the best specifications for the predictive models, the investor forecasts and generates sample draws from the one-step-ahead conditional return distributions, conditioning on the previous realized returns and predictors, and use the sample draws to find the optimal portfolio weights as described in Section 1. To measure the performance of the portfolios formed in this manner, we use several performance measures including the Sharpe ratio, certainty equivalent rate of return (CER), and a measure proposed by Graham and Harvey (1997) (henceforth $\mathcal{GH}2$). Note that we compute and compare *ex post* CER or sample CER instead of *ex ante* CER used in studies such as Kandel and Stambaugh (1996), Pástor (2000), and Pástor and Stambaugh (2000). The sample CERs are calculated by taking the average of the realized utilities over the period considered;

$$\mu(W_0(1 + r_{ce})) = \frac{1}{T} \sum_{t=1}^T \mu(W_0(1 + r_{p,t})), \quad (10)$$

where r_{ce} is the sample CER, $r_{p,t}$ is the realized portfolio return at time t , and $\mu(\cdot)$ is the utility function. The rationale is that if the predictive model is a good description of the true underlying data-generating process, the investor who allocates her wealth each period according to the predictive model should be better off on average judged by the realized returns. $\mathcal{GH}2$ is directly related to the Sharpe ratio but also quantifies the outperformance; it is the abnormal return that the measured portfolio would have earned if it had the same risk as the market portfolio. In other words, we first lever up or down the measured portfolio with 1-month T-bill (riskfree asset), so that the levered portfolio has the same risk (volatility) as the market portfolio. We then compare the average return of the levered portfolio with that of the market portfolio.¹³

¹³ $\mathcal{GH}2$ is similar to the well-known $\mathcal{M}2$ measure except that the riskfree rate is not constant over time.

It amounts to find the weight ω to solve the following problem:

$$V_m = Var(\omega r_{pt} + (1 - \omega)r_{ft}) = \omega^2 V_p + (1 - \omega)^2 V_f + 2\omega(1 - \omega)Cor(r_{pt}, r_{ft}), \quad (11)$$

where V_m , V_p , and V_f are the variances of the market returns, managed portfolio returns, and riskfree rates, respectively. $\mathcal{GH2}$ is given as

$$\mathcal{GH2} = \omega \bar{r}_p + (1 - \omega)\bar{r}_f - \bar{r}_m, \quad (12)$$

where $\bar{r}_p$, $\bar{r}_f$, and $\bar{r}_m$ are average returns on the managed portfolio, 1-month T-bill, and the market portfolio respectively.

4.1 In-Sample Performance Analysis

In the in-sample tests, we use the whole sample period from 1947:1 (or 1953:4 for the term spread) to 1998:12 as the estimation period and testing period. Table 3 reports in-sample performance results. We report results for the case where no constraint is imposed on the portfolio weights and the case where no short sale is allowed. As benchmarks, we consider two strategies: fixed-weight strategy where the returns are perceived IID and therefore the weight is always the same, and buy-and-hold passive strategy where the weight is determined at the beginning of the period and no rebalance of portfolio is required thereafter.

The most prominent result from Table 3 is that predictive models outperform the benchmarks across the board. T-bill yield, default spread, term spread, and various combinations of the three variables all yield superior performance with each predictive model except default spread with OPT and NLNP where the performance is close to that of the benchmarks. Imposing the no-short-sale constraint only slightly reduces the performance. The only exception is the dividend yield, which does not outperform the benchmarks except its VAR model. This is in sharp contrast to other predictive variables and suggests that dividend yield does not seem to have any predictive power over the market returns, which is not surprising given the weak correlation between the dividend yield and the market returns. In addition, T-bill yield seems to yield the best performance and thus to be the most powerful predictor, followed by term spread and default spread. On the other hand, term spread and default spread seem to complement the T-bill yield as the combinations of each one with the T-bill yield produce even stronger performance. Furthermore, to our surprise, the optimal SNP model does not perform well, and in fact it often yields the worst performance. By contrast, the simplest model in the group, the VAR model, yield the strongest performance in many

cases, while in other cases the best performance is obtained by the overfitted NLNP model. Finally, the in-sample performance is very robust - different performance measures give rise to the same results.

4.2 Out-of-Sample Performance Analysis

The in-sample tests report strong outperformance for the predictive variables and predictive models. Because the estimation period is the same as the testing period in the in-sample tests, however, the in-sample may not be relevant to the real-time performance. To evaluate the real-time performance of the predictive models and predictive variables, we shall assume that the investor only has information about historical values of the market excess returns and predictive variables.

To perform out-of-sample tests, we use the last 20 years in the sample period as the testing period, i.e. from 1979:1 to 1998:12, and the rest as the estimation period. The period from 1979 to 1998 is an interesting period as it contains some recession periods (1980:1–1980:7, 1981:7–1982:11, and 1990:7–1991:3) and the longest boom period in 1990’s. Table 4 reports the out-of-sample results. In sharp contrast to the in-sample results, no predictive models yield performance superior to the benchmarks regardless of the combinations of predictive variables present when no constraint on the portfolio weights is imposed. In other words, every predictive model of every combination of the predictive variables significantly underperforms the benchmarks with negative measures of the CER and $\mathcal{GH}2$, and in some cases the underperformance is very severe. For example, the OPT model with the term spread has a Sharpe ratio of 0.05, a $\mathcal{GH}2$ of -5.13%, and a CER of -163.1%. However, the biggest surprise is the T-bill yield, which is the strongest predictor in the in-sample analysis, yields the worst performance.

On the other hand, the most interesting result in this table is that imposing the no-short-sale constraint substantially improves the performance of the predictive models, which is also in contrast to the in-sample results where imposing the constraint slightly reduce the performance. However, even with the help of the constraint, the predictive models with a single predictor still underperform the benchmarks except the VAR-GARCH model of the term spread. With two or more predictive variables, the situation is better, but still only selected models outperform the benchmarks. For example, with T-bill yield and default spread combined, the VAR and NLNP models yield superior performance, but the VAR-GARCH and OPT models underperform the benchmarks. However, the VAR-GARCH model seems to perform the best in four out of eight cases of no-short-sale, whereas the OPT model often ranks the last. Comparing the case of no constraint to the one with no-short-sell constraint, the former case has much higher volatility than the later

case, which likely explains the underperformance of the former case.

4.3 Recursive Estimation and Performance Tests

From Table 1 and Table 2, we know that the relations between the market returns and predictive variables are unstable over time, and the best specifications may change over different periods. Given that, the investor would be better off repeating the specification search and estimation again after certain periods to capture the possible structural and parametric changes. In this subsection, we consider recursive estimation where the estimation is repeated every five years with an expanding window of periods. As in the out-of-sample analysis, the investor searches for the best specifications using data from the estimation period (1947:1–1979:12), generates sample draws from one-step-ahead predictions of the return distributions, calculates the optimal weights using the sample draws, and allocates between the market portfolio and riskfree asset accordingly. After five years, however, the investor repeats the search for the best specification using data from the estimation period plus the most recent five years (expanding window), uses the new specification and parameter estimates to choose the optimal portfolio weights. The investor repeats the same procedure every five years.

Table 5 reports the performance results. Similar to the out-of-sample results, when no constraint is imposed, no predictive model of any combination of predictive variables outperforms the benchmarks, which includes an expanding strategy where the weights are fixed for five years and then are updated using returns up to that moment. Similarly, restricting the weights between zero and one substantially improves the performance, but only a few combinations of predictive variables manage to outperform the benchmarks such as TBYD, TBYD and DFSY combination, and TBYD and TRSD combination. Compared to the out-of-sample results, the recursive estimation results are slightly better in general but worse in a few cases. The most notable difference, however, is the result of T-bill yield, which underperforms the benchmarks in the out-of-sample analysis but outperforms the benchmarks in the recursive estimation analysis when the weights are restricted. The VAR-GARCH model also performs best in four out of eight cases.

4.4 Market Timing

Results from both the out-of-sample analysis and recursive estimation show that some predictive variables such as T-bill yield or combinations of T-bill yield with other predictive variables yield superior performance

when the portfolio weights are restricted, whereas no superior performance can be found when the portfolio weights are not restricted. It is worth noting that when the portfolio weights are restricted, at least 60% of the weights are changes to either 0 or 1. This means that more often than not, the optimal weights computed from the predictive models may not be correct, and better performance is obtained if the portfolio weights are restricted to either pure cash position or pure equity position. The evidence suggests that we should make a finer distinction about the predictive ability of the predictive variables: ability to predict the magnitude of the market expected excess return or the ability to predict the sign of the market expected excess return. No predictive variables have the ability to predict the magnitude of the market expected excess return out of sample, but a few variables such as T-bill yield have the ability to predict the sign or direction of changes. In other words, T-bill yield can be used to predict whether the market will go up or down, but it can not tell us by how much the market will move up or down. To further support this finding, we examine the performance of switching portfolios. By construction, switching portfolios switch from the all-equity position to the pure-cash position (riskfree asset) or vice versa depending on whether the forecasted expected excess returns are positive or negative. Therefore, switching portfolios only time the direction of market movement.

Table 6 reports the performance of the switching portfolios. Interestingly, all combinations except dividend yield and default spread generate superior performance in at least one predictive model, and most outperform in all predictive models. In addition, the performance is stronger than that in the recursive estimation analysis. For example, T-bill yield generates higher Sharpe ratios and $\mathcal{GH}2$ measures than those in the recursive estimation analysis, and term spread outperforms the benchmarks in every predictive model, whereas it underperforms the benchmarks in the recursive estimation analysis. Results in this table indeed suggest that the investor will be better off not predicting the magnitude of the market movement at all but focusing on the direction of movement.

4.5 Transaction Costs

To investigate the real-time performance, it is necessary to incorporate appropriate transaction costs into the portfolio choice problem. We consider three levels of transaction costs: 0.25%, 0.50%, and 1.00%, representing low, medium, and high transaction costs. Table 7 reports the performance of two combinations of predictive variables: TBYD & DFSD combination, and TBYD & TRSD combination. Results of other combinations are similar and thus are omitted in the interest of conciseness. With the low level of transaction

cost, there is virtually no impact on the performance. In some cases, the performance is actually slightly better. This is not surprising given that the investor incorporates transaction costs into her objective function and optimally chooses the portfolio weights. Similar results are obtained for the medium level of transaction cost, although it starts to show the impact, especially with the no-short-sale case. For the high level of transaction cost, however, the effect is apparent. For example, the combination of TBYD & DFSD yields superior performance when the portfolio weights are restricted, but underperforms the benchmarks when transaction costs are high. Interestingly, there seems to be no effect on the performance when no restriction is imposed on the portfolio weights.

4.6 Subperiod Portfolio Performance

Since Table 1 and Table 2 suggest that the relation between the market returns and predictive variables are unstable over time, one needs to be careful about interpreting the results. For example, the results may be specific to the testing period we consider. Therefore, we conduct subperiod performance analysis to examine: 1) if the results are specific to the period we choose, and 2) if there are any interesting dynamics of the managed portfolios. Table 8 reports the results for T-bill yield in the case of no short sale. Other combinations of predictive variables yield similar results. Two different sets of results are shown: Panel A shows the period-by-period performance, while Panel B shows the accumulative performance. In Panel A, the top rows report the performance of the expanding benchmark portfolio. In the first five years (1979–1983), the performance of the benchmark is the lowest with a Sharpe ratio of 0.05 because of recessions in this period (1980 and 1982). The performance improves in the second and third five-year periods, and reach the highest in the last five years (1994–1998) with a Sharpe ratio of 1.10. Similar pattern in Sharpe ratio shows up in the VAR model of T-bill yield, but the other three models display a different dynamics in Sharpe ratio – the lowest performance happens in the third 5-year period. Nevertheless, a common theme from all the predictive models is that the T-bill yield outperforms the benchmark in the first 10 years when the market performs poorly but underperforms the benchmark in the second 10 years when the market performs well. This is clearly shown by the last two columns of Panel A, which report the differences in $\mathcal{GH}2$ and r_{ce} respectively. For example, the VAR model ranks the highest in the last two periods but ranks the last in the first two periods, whereas the OPT model ranks the first in the first two periods but ranks the last in the last two periods.

Despite that the predictive models of T-bill yield underperform the benchmarks in the last two five-year

periods, the accumulative performance of all the predictive models are superior to that of the benchmark portfolio in every subperiod. Results from this table shows that the accumulative performance is robust to different time periods,¹⁴ but the performance is not stable. The outperformance decreases in the last ten years of the sample periods. Results from Table 8 suggest that in periods of bad time, the investor should employ the predictive models to guide her portfolio choice, or better yet employ the switching strategy, whereas in periods of good time, the investor is better off disregarding any information in the predictive variables.

4.7 Robustness Analysis

We have shown that when the predictive models and combinations of predictive variables yield superior performance, the outperformance is robust to time periods and appropriate transaction costs. In this subsection, we further explore the robustness of the outperformance in other dimensions. First, we change the number of years between re-estimation in the recursive estimation analysis from five years to two years. Second, we change the relative risk aversion coefficient from four to ten. Third, we increase the investment horizon from one month to three months or six months. In all three cases, the results (not reported) are qualitatively similar.

Finally, we change the investor's preference from power utility to mean-variance utility. Table 9 reports the portfolio performance under this new preference.¹⁵ Interestingly, the results are similar to those of the power utility. For example, with the portfolio weights restricted, TBYD and the combination of TBYD & DFSD outperform the benchmarks with every predictive model, and the combination of TBYD & TRSD and combination of TBYD, DFSD, & TRSD have two or three predictive models outperforming the benchmarks. However, the performance numbers are slightly lower. For example, with the mean-variance preference, T-bill yield generates Sharpe ratios around 0.49, and $\mathcal{GH}2$ ranging from 1.22 to 1.67, whereas with the power utility, T-bill yield generates Sharpe ratios from 0.51 to 0.56 and $\mathcal{GH}2$ from 1.89 to 2.59. On the other hand, results of no constraint are stronger as the VAR model of TRSD and the combination of TBYD & DFSD outperform the benchmarks, but in most cases, the predictive models still underperform the benchmarks.

¹⁴Of course, the robustness of the outperformance depends on the starting time. As shown in Panel A, if the starting time is 1989, then outperformance changes to underperformance because the predictive models perform worse than the benchmark in the subsequent 10-year period.

¹⁵Following the convention, we use simple returns with mean-variance preference, in which case all benchmark strategies yield the same Sharpe ratio.

5 Conclusion

Extensive empirical evidence suggests that the stock returns are predictable to some extent by many economic variables. Subsequently, the implication of return predictability on investor's portfolio choice problem has been extensively studied. Although studies generally find significant impact on how investors allocate their assets, the actual performance of the portfolios constructed based on return predictability is still under investigation, and the question whether investors can benefit from employing the predictive models to guide their asset allocation decisions is still subject to debate. Furthermore, much of the attention has been focused on the predictive variables, but the specification of the underlying data-generating process has received little attention. Almost all the studies employ a VAR model or a further simplified linear regression model, and it is likely that these kinds of simple models are mis-specified. In this paper, we evaluate the real-time economic significance of return predictability by taking a more careful approach that guards against possible model mis-specifications. In particular, we consider an array of predictive models from the simplest VAR model to the nonlinear nonparametric model and select the best model specifications using statistical criteria.

We conduct extensive in-sample as well as out-of-sample analysis to evaluate the portfolio performance of the predictive models, using the widely accepted predictive variables such as the dividend yield, T-bill yield, term spread, and default spread. First, while we find strong in-sample evidence of predictability, the evidence of out-of-sample predictability is fairly weak. In particular, we find that the predictive variables yield superior portfolio performance only when no short-selling is allowed or when a switching strategy is used, which suggests that predictive variables may be able to predict the sign but not the magnitude of the market expected returns. In other words, the predictive variables can only be used to signal either a buy or sell signal, and nothing more. A possible explanation is related to the fact that, as pointed out by Merton (1980), there is too much noise in the observed returns to accurately estimate the expected returns even if the predictive relation holds. Indeed, Torous and Valkanov (2000) argue that even if returns are predictable, the noise in the predictive regression may overwhelm the signal of the conditional variables. On the other hand, Christoffersen and Diebold (2003) show analytically that the sign of asset returns are predictable and the predictability depends on the predictability of return volatility. Our finding may be related to their findings. Alternatively, it is likely that models considered here are mis-specified despite our effort to use the flexible seminonparametric specification. Other flexible specifications such as one that includes latent states may be worth considering.

Second, we investigate the subperiod performance of the outperforming predictive models and find that the predictive models yield superior performance when the market conditions are poor and yield inferior performance when the market conditions are good. Our finding suggests that the predictive variables are useful precisely when they are needed the most. On the other hand, when the market is booming, investors may be better off following simple strategies such as expanding window or buy-and-hold strategy.

Third, it seems that the VAR-GARCH model performs the best, whereas the best statistical model performs the worst. On the other hand, the overfitted nonlinear nonparametric model also performs well, sometimes better than the VAR-GARCH model. These results have several implications. First, it suggests that it is important to choose the right criterion; for the purpose of asset allocation, one should use an appropriate financial criterion. In similar spirit, Aït-Sahalia and Brandt (2001) argue that one should choose predictive variables that are important for optimal portfolio weights, rather than for moments of the conditional distribution of returns, when considering portfolio choice problems. Pesaran and Timmermann (1995) also use financial criteria along with the statistical criteria for model selection. Second, in-sample inference may not be relevant to out-of-sample performance as the model is chosen according to in-sample criteria. This is consistent with the findings of Bossaerts and Hillion (1999). Finally, our results suggest that it is necessary to model conditional heteroscedasticity in return predictability.

Our analysis can be extended in a number of interesting directions. First, we do not consider dynamic hedging demands. A dynamic strategy that hedges future changes in the investment opportunity set may perform quite differently. Second, we assume perfect foresight and ignore estimation risk. We suspect that taking into account parameter uncertainty and learning may reduce the performance, and therefore it remains to see that if the outperformance is robust to these uncertainties. Third, we examine four economic variables that are believed to be powerful predictors, but there are other potentially powerful predictive variables that may be worth investigating. Fourth, we consider predictability at monthly level, it may be of interest considering predictability at quarterly and even annual levels. For example, Lettau and Ludvigson (2001) find that the consumption-wealth ratio is a very powerful predictive variable, which is only available at quarterly and annual level. Fifth, since different predictive models perform differently in different periods, it may be beneficial for investors to use a model averaging approach. Unlike Avramov (2002) and Cremers (2002) who average over a set of predictors, a set of data-generating processes such as the ones considered here would be averaged. Finally, our results suggest that the predictive relations seem unstable, and we employ a strategy that re-estimates the model periodically. But an alternative strategy is to explicitly model

time-varying parameters or structural breaks. It is well known that regime shift happens with T-bill yield. Rapach and Wohar (2004) find significant evidence of structural breaks in seven of eight predictive regressions of S&P 500 returns and three of eight in CRSP equal-weighted returns. Pesaran and Timmermann (2002) find that a linear predictive model that incorporates structural breaks has improved out-of-sample statistical forecasting power.

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Table 1: Descriptive Statistics of Data

Panel A of this table shows the descriptive statistics for the continuously compounded excess return (annualized in percentage) on the S&P 500 composite index (EXRN), continuously compounded return on 1-month T-bill (riskfree return), dividend yield (DVYD), and 3-month Treasury bill yield (TBYD), default spread (DFSD), and term spread (TRSD). For all the variables except the term spread, the data is sampled monthly from January 1947 through December 1998, with a total of 624 observations. For TRSD, the data is only available from April, 1953. The whole sample period is divided into two subperiods: the first subperiod is from 1947:1 to 1978:12 (or 1953:4 - 1978:12 for TRSD); the second subperiod is from 1979:1 to 1998:12. Panel B shows the correlations of the predictive variables and excess return for the whole sample period and the two subperiods.

Panel A: Descriptive Statistics

	Mean	StdDev	Skewness	Kurtosis	JB	Autocorrelations			
						ρ_1	ρ_3	ρ_6	ρ_{12}
EXRN	3.654	14.267	-0.640	5.747	238.9	0.022	0.012	-0.059	0.042
RFT1M	4.782	0.869	1.045	4.527	174.2	0.955	0.922	0.877	0.808
DVYD	3.950	1.211	0.460	2.585	26.6	0.989	0.962	0.919	0.835
TBYD	4.954	2.998	0.973	4.208	136.5	0.988	0.952	0.909	0.847
DFSD	0.908	0.426	1.532	5.345	387.4	0.976	0.918	0.850	0.720
TRSD	0.718	0.990	-0.137	3.463	6.6	0.960	0.841	0.710	0.517
EXRN(47:1-78:12)	2.296	13.638	-0.261	3.725	127.7	0.034	0.072	-0.075	0.085
EXRN(79:1-98:12)	5.825	15.228	-1.099	4.828	284.6	0.004	-0.061	-0.041	-0.037

Panel B: Correlations

EXRN	DVYD	TBYD	DFSD	TRSD	EXRN	DVYD	TBYD	DFSD
1953:4-1998:12					1947:1-1998:12			
1.000	-0.069	-0.153	0.053	0.165	1.000	-0.030	-0.140	0.043
	1.000	0.409	0.497	-0.191		1.000	0.011	0.228
		1.000	0.643	-0.408			1.000	0.664
			1.000	0.105				1.000
				1.000				
1953:4-1978:12					1947:1-1978:12			
1.000	-0.005	-0.266	0.095	0.280	1.000	0.045	-0.225	0.074
	1.000	-0.025	0.150	0.111		1.000	-0.425	-0.077
		1.000	0.343	-0.508			1.000	0.399
			1.000	0.386				1.000
				1.000				
1979:1-1998:12								
1.000	-0.115	-0.167	-0.005	0.080				
	1.000	0.807	0.801	-0.328				
		1.000	0.681	-0.703				
			1.000	-0.179				
				1.000				

Table 2: Optimal Predictive Models

This table reports for each combination of predictive variables the best predictive model and the various model selection criteria used in each category. Panel A reports the in-sample predictive models, while Panel B reports the out-of-sample predictive models. Obj is the value of the objective function and is defined as $Obj = -(\frac{1}{n} \sum_{t=1}^n \log[f(y_t|x_{t-1}, \theta)])$, and BIC, AIC, and HQ are the Schwarz Bayesian Information Criterion, the Akaike Information criterion, and the Hannan-Quinn criterion, respectively. We use the following notation to denote the various models, " $(m_1, m_2, m_3, m_4, m_5)$ ", where m_1 denotes the order of vector autoregression, m_2 and m_3 denote the order of GARCH (GARCH(m_2, m_3)), m_4 denotes the order of the Hermite polynomial expansion, and m_5 is the order of polynomial in x for the coefficients in the Hermite polynomial.

Optimal Predictive Models and Model Selection Criteria																				
	Model	Obj	BIC	AIC	HQ	Model	Obj	BIC	AIC	HQ	Model	Obj	BIC	AIC	HQ	Model	Obj	BIC	AIC	HQ
Panel A: In-Sample Estimation (Full Sample Period)																				
	DVYD					TBYD					DFSD					TBYD & DFSD				
VAR	(1,0,0,0,0)	0.40	0.45	0.41	0.43	(2,0,0,0,0)	0.85	0.92	0.87	0.89	(2,0,0,0,0)	1.29	1.36	1.31	1.33	(2,0,0,0,0)	0.61	0.75	0.65	0.69
GARCH	(2,2,1,0,0)	0.19	0.28	0.22	0.24	(2,2,1,0,0)	0.20	0.29	0.23	0.25	(2,1,1,0,0)	0.85	0.93	0.88	0.90	(2,1,1,0,0)	-0.30	-0.14	-0.25	-0.21
OPT	(2,2,1,4,0)	0.06	0.20	0.10	0.14	(2,2,1,4,0)	0.13	0.27	0.17	0.21	(2,1,1,4,0)	0.76	0.89	0.80	0.83	(2,1,1,4,0)	-0.42	-0.20	-0.35	-0.29
NLNP	(2,2,1,4,1)	-0.02	0.25	0.06	0.14	(2,2,1,4,1)	0.11	0.35	0.18	0.25	(2,1,1,4,1)	0.73	0.95	0.80	0.86	(2,1,1,4,1)	-0.50	-0.07	-0.36	-0.25
	TRSD					TBYD & TRSD					TRSD & DFSD					TBYD & TRSD & DFSD				
VAR	(2,0,0,0,0)	1.50	1.58	1.53	1.55	(2,0,0,0,0)	0.42	0.57	0.47	0.51	(2,0,0,0,0)	1.34	1.50	1.39	1.43	(2,0,0,0,0)	0.18	0.45	0.27	0.34
GARCH	(2,1,1,0,0)	1.21	1.30	1.24	1.26	(2,1,1,0,0)	0.03	0.21	0.09	0.14	(2,1,1,0,0)	0.68	0.87	0.74	0.79	(2,1,1,0,0)	-0.50	-0.21	-0.41	-0.33
OPT	(2,1,1,4,0)	1.15	1.29	1.19	1.23	(2,1,1,4,0)	-0.11	0.16	-0.02	0.05	(2,1,1,4,0)	0.57	0.82	0.65	0.72	(2,1,1,4,0)	-0.69	-0.26	-0.55	-0.44
NLNP	(2,1,1,4,1)	1.09	1.33	1.17	1.23	(2,1,1,4,1)	-0.23	0.32	-0.05	0.09	(2,1,1,4,1)	0.47	0.95	0.62	0.75	(2,1,1,4,1)	-0.85	0.12	-0.54	-0.28
Panel B: Out-Of-Sample Estimation (Estimation Period)																				
	DVYD					TBYD					DFSD					TBYD & DFSD				
VAR	(1,0,0,0,0)	0.47	0.54	0.49	0.51	(2,0,0,0,0)	0.76	0.86	0.79	0.82	(2,0,0,0,0)	1.25	1.35	1.28	1.31	(2,0,0,0,0)	0.53	0.75	0.61	0.66
GARCH	(1,1,1,0,0)	0.27	0.36	0.30	0.32	(2,1,1,0,0)	0.31	0.45	0.36	0.39	(2,1,1,0,0)	0.96	1.09	1.01	1.04	(2,2,1,0,0)	-0.02	0.24	0.07	0.13
OPT	(1,1,1,4,0)	0.16	0.32	0.22	0.26	(2,1,1,4,0)	0.24	0.45	0.31	0.37	(2,1,1,4,0)	0.90	1.09	0.96	1.01	(2,2,1,4,0)	-0.22	0.15	-0.10	0.00
NLNP	(1,1,1,4,1)	0.09	0.39	0.20	0.27	(2,1,1,4,1)	0.19	0.56	0.32	0.41	(2,1,1,4,1)	0.89	1.10	0.96	1.02	(2,2,1,4,1)	-0.37	0.38	-0.12	0.08
	TRSD					TBYD & TRSD					TRSD & DFSD					TBYD & TRSD & DFSD				
VAR	(2,0,0,0,0)	1.54	1.66	1.58	1.61	(2,0,0,0,0)	0.52	0.78	0.61	0.68	(1,0,0,0,0)	1.40	1.57	1.46	1.50	(1,0,0,0,0)	0.39	0.68	0.49	0.57
GARCH	(2,1,1,0,0)	1.39	1.54	1.44	1.48	(2,1,1,0,0)	0.36	0.66	0.47	0.54	(1,1,1,0,0)	0.96	1.17	1.03	1.09	(1,1,1,0,0)	-0.04	0.29	0.08	0.16
OPT	(2,1,1,4,0)	1.32	1.55	1.40	1.46	(2,1,1,4,0)	0.24	0.66	0.39	0.49	(1,1,1,4,0)	0.84	1.17	0.96	1.04	(1,1,1,4,0)	-0.22	0.27	-0.05	0.08
NLNP	(2,1,1,4,1)	1.27	1.67	1.41	1.52	(2,1,1,4,1)	0.10	0.88	0.37	0.58	(1,1,1,4,1)	0.71	1.40	0.95	1.13	(1,1,1,4,1)	-0.46	0.67	-0.07	0.23

Table 3: In-Sample Predictive Variable Performance

Predictive sample draws $\tilde{\mathbf{y}}_{t+1|t}$ are generated at each month t from the one-step-ahead conditional distributions of the predictive models, conditioned on the observed data $\mathbf{y}_t$. The predictive sample draws of the excess returns are then used to solve the portfolio optimization problem at each month t . The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns, standard deviations, and three performance measures, Sharpe ratio ($\mathcal{SR}$), Graham-Harvey measure ($\mathcal{GH2}$), and CER (r_{ce}), are reported for each predictive variable and model combination. The number of sample draws at each period t is 50000.

In-Sample Portfolio Performance Tests																				
	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$
Benchmarks																				
Market	8.44	14.17	0.26		5.35															
Fixed	7.03	7.69	0.29	0.48	6.14															
Passive	7.50	10.59	0.25	-0.03	5.78															
	DVYD					TBYD					DFSD					TBYD & DFSD				
	No Constraint					No Constraint					No Constraint					No Constraint				
VAR	7.78	8.46	0.34	1.18	6.71	15.54	18.65	0.58	4.40	10.29	8.73	11.05	0.35	1.26	6.92	20.67	23.09	0.69	5.98	13.04
GARCH	8.33	15.52	0.22	-0.55	4.72	14.86	17.71	0.57	4.29	10.10	8.71	12.95	0.30	0.50	6.18	19.11	20.09	0.71	6.34	13.28
OPT	7.42	15.04	0.17	-1.30	4.00	13.90	16.98	0.53	3.83	9.47	8.28	14.22	0.24	-0.30	5.16	18.39	19.37	0.70	6.18	13.11
NLNP	7.52	10.61	0.25	-0.16	5.86	13.27	15.50	0.54	3.99	9.60	7.88	11.11	0.27	0.15	6.04	19.28	20.26	0.71	6.38	13.34
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale				
VAR	7.65	8.21	0.34	1.10	6.64	10.36	10.03	0.55	4.07	8.85	8.09	9.03	0.36	1.37	6.88	11.26	9.82	0.65	5.55	9.83
GARCH	7.52	10.11	0.26	0.02	5.99	9.91	9.42	0.53	3.90	8.57	8.33	10.92	0.32	0.79	6.52	11.31	9.95	0.65	5.49	9.83
OPT	7.42	10.21	0.25	-0.17	5.85	9.92	9.95	0.51	3.51	8.42	7.53	10.45	0.26	-0.09	5.84	11.00	10.02	0.61	4.99	9.50
NLNP	7.13	8.39	0.27	0.12	6.08	9.77	9.75	0.50	3.43	8.33	7.56	9.72	0.28	0.22	6.12	10.73	9.46	0.62	5.09	9.39
	TRSD					TBYD & TRSD					TRSD & DFSD					TBYD & TRSD & DFSD				
	No Constraint					No Constraint					No Constraint					No Constraint				
VAR	13.13	14.13	0.55	4.62	10.07	19.49	21.24	0.67	6.26	13.06	14.86	17.33	0.55	4.58	10.25	23.67	25.35	0.73	7.07	14.82
GARCH	11.24	11.52	0.51	4.06	9.21	17.98	20.36	0.62	5.62	11.90	13.26	15.49	0.51	4.04	9.56	21.32	22.57	0.71	6.86	14.15
OPT	10.03	11.19	0.42	2.72	8.05	16.30	18.49	0.59	5.21	11.18	10.83	15.36	0.36	1.83	7.05	19.55	19.99	0.71	6.90	13.98
NLNP	13.19	15.05	0.52	4.20	9.41	19.71	18.36	0.78	7.96	14.84	16.56	18.31	0.61	5.51	11.04	22.57	20.95	0.82	8.52	16.26
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale				
VAR	9.89	9.11	0.50	3.88	8.59	12.39	9.78	0.72	7.07	10.98	9.76	9.60	0.46	3.31	8.33	12.05	9.56	0.70	6.79	10.70
GARCH	9.70	9.01	0.48	3.66	8.43	11.18	10.10	0.58	5.01	9.66	10.16	10.27	0.47	3.44	8.53	11.56	9.88	0.63	5.75	10.12
OPT	8.76	8.96	0.38	2.17	7.49	10.51	10.41	0.49	3.83	8.88	8.72	9.55	0.35	1.78	7.30	11.39	9.56	0.63	5.80	10.04
NLNP	10.61	9.96	0.53	4.30	9.07	11.60	9.74	0.64	5.93	10.18	11.26	10.37	0.57	4.90	9.60	11.89	9.60	0.68	6.50	10.51

Table 4: Out-Of-Sample Predictive Variable Performance

Predictive sample draws of the excess returns $\tilde{\mathbf{r}}_{t+1|t}$ are generated at each month t from the one-step-ahead conditional distributions of the predictive models, conditioned on the observed out-of-sample data $\mathbf{y}_t$, and are then used to solve the portfolio optimization problem at each month t . The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns, standard deviations, and three performance measures, Sharpe ratio ($\mathcal{SR}$), Graham-Harvey measure ($\mathcal{GH2}$), and CER (r_{ce}), are reported for each predictive variable and model combination. The number of sample draws at each period t is 50000.

Out-Of-Sample Portfolio Performance Tests																					
	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	
Benchmarks																					
Market	12.74	15.10	0.38		9.18																
Fixed	9.63	6.20	0.43	0.75	9.05																
Passive	10.18	7.47	0.43	0.74	9.32																
	DVYD					TBYD					DFSD					TBYD & DFSD					
	No Constraint					No Constraint					No Constraint					No Constraint					
	VAR	4.13	13.90	-0.20	-8.86	1.32	5.88	53.03	-0.02	-6.12	-91.33	9.82	16.43	0.17	-3.16	5.70	12.49	45.77	0.12	-4.00	-26.00
	GARCH	8.10	12.81	0.09	-4.43	5.65	1.85	47.34	-0.11	-7.43	-57.32	10.10	14.71	0.21	-2.56	6.80	8.75	34.59	0.05	-5.03	-12.46
	OPT	-1.73	22.77	-0.38	-11.56	-9.45	-30.8	136.7	-0.28	-9.99	-3804	9.69	16.00	0.17	-3.21	5.58	6.53	39.11	-0.01	-5.97	-20.26
	NLNP	3.12	14.90	-0.26	-9.68	-0.19	3.46	31.54	-0.11	-7.47	-12.02	9.76	14.88	0.19	-2.94	6.21	6.46	32.87	-0.01	-6.03	-10.07
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale					
	VAR	8.01	7.71	0.14	-3.69	7.13	8.74	5.09	0.36	-0.39	8.36	10.16	11.20	0.29	-1.47	8.26	11.23	7.04	0.61	3.45	10.51
	GARCH	8.89	9.29	0.21	-2.63	7.61	8.05	3.17	0.36	-0.34	7.90	10.70	11.66	0.32	-0.94	8.63	8.76	4.78	0.39	0.04	8.42
	OPT	7.51	8.16	0.07	-4.74	6.52	8.13	3.77	0.32	-0.93	7.92	9.13	11.03	0.20	-2.80	7.16	9.05	5.11	0.42	0.51	8.66
NLNP	7.76	6.06	0.14	-3.73	7.22	7.75	2.74	0.31	-1.10	7.64	9.55	10.67	0.24	-2.11	7.70	10.05	6.20	0.51	1.83	9.48	
	TRSD					TBYD & TRSD					TRSD & DFSD					TBYD & TRSD & DFSD					
	No Constraint					No Constraint					No Constraint					No Constraint					
	VAR	17.16	45.92	0.22	-2.47	-30.15	15.74	61.85	0.14	-3.68	-193.1	15.02	40.35	0.20	-2.80	-26.40	11.02	38.68	0.11	-4.23	-11.14
	GARCH	15.33	41.11	0.20	-2.75	-32.79	14.28	40.53	0.18	-3.10	-14.81	13.20	42.91	0.15	-3.62	-37.55	7.04	40.08	0.00	-5.78	-20.70
	OPT	9.21	49.93	0.05	-5.13	-163.1	7.35	56.45	0.01	-5.71	-157.3	12.35	37.81	0.14	-3.67	-35.36	2.27	44.15	-0.11	-7.40	-30.54
	NLNP	9.58	52.85	0.05	-5.07	-86.42	8.09	44.99	0.03	-5.43	-37.76	13.45	38.40	0.17	-3.25	-34.96	4.54	49.05	-0.05	-6.55	-35.88
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale					
	VAR	12.07	11.84	0.43	0.74	9.84	11.50	7.85	0.58	2.99	10.58	12.46	11.76	0.47	1.29	10.26	9.80	6.50	0.44	0.90	9.18
	GARCH	12.49	12.21	0.45	1.06	10.12	10.65	7.26	0.51	1.95	9.86	12.54	11.57	0.48	1.51	10.39	9.56	5.92	0.45	0.94	9.05
	OPT	10.84	11.22	0.35	-0.55	8.81	9.59	6.60	0.41	0.31	8.94	12.40	11.37	0.48	1.46	10.33	9.17	4.85	0.47	1.23	8.84
NLNP	11.79	11.85	0.41	0.38	9.54	9.09	6.87	0.32	-1.04	8.38	12.14	12.42	0.42	0.52	9.70	10.83	5.78	0.67	4.38	10.35	

Table 5: Out-Of-Sample Predictive Variable Expanding Performance

Predictive sample draws of the excess returns $\tilde{\mathbf{r}}_{t+1|t}$ are generated at each month t from the one-step-ahead conditional distributions of the predictive models, conditioned on the observed out-of-sample data $\mathbf{y}_t$. The predictive models are re-estimated every 5 years. The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns, standard deviations, and three performance measures, Sharpe ratio ($\mathcal{SR}$), Graham-Harvey measure ($\mathcal{GH2}$), and CER (r_{ce}), are reported for each predictive variable and model combination. The number of sample draws at each period t is 50000.

Out-Of-Sample 5-year Expanding Portfolio Performance Tests																					
	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	
Benchmarks																					
Market	12.74	15.10	0.38		9.18																
Passive	10.18	7.47	0.43	0.74	9.32																
Expand	9.96	6.06	0.45	0.95	9.40																
30	DVYD					TBYD					DFSD					TBYD & DFSD					
	No Constraint					No Constraint					No Constraint					No Constraint					
	VAR	5.25	12.33	-0.14	-7.87	3.04	13.78	46.63	0.15	-3.62	-75.30	7.65	15.40	0.05	-5.11	4.02	18.15	41.86	0.27	-1.80	-15.53
	GARCH	7.54	12.66	0.05	-5.09	5.14	9.88	38.67	0.08	-4.67	-35.30	10.29	14.41	0.23	-2.29	7.10	14.33	29.76	0.25	-2.08	-0.58
	OPT	4.54	16.22	-0.15	-8.04	0.63	-21.45	133.5	-0.21	-9.04	-3804	10.27	15.57	0.21	-2.57	6.39	9.76	36.62	0.08	-4.66	-13.61
	NLNP	0.62	16.57	-0.38	-11.56	-3.45	10.60	23.69	0.16	-3.48	1.41	10.78	11.44	0.34	-0.74	8.86	9.05	39.59	0.05	-5.02	-38.06
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale					
	VAR	7.78	7.44	0.11	-4.09	6.95	10.38	6.31	0.55	2.45	9.78	7.85	10.31	0.09	-4.47	6.21	11.89	7.60	0.65	4.07	11.05
	GARCH	8.69	8.95	0.20	-2.84	7.50	10.23	5.94	0.56	2.59	9.70	11.00	11.05	0.37	-0.26	9.15	11.52	7.35	0.62	3.63	10.72
	OPT	7.64	8.05	0.09	-4.48	6.68	10.21	6.24	0.53	2.16	9.62	9.54	11.39	0.23	-2.36	7.46	10.16	6.35	0.51	1.89	9.58
NLNP	8.09	6.39	0.18	-3.06	7.49	9.80	5.65	0.51	1.89	9.32	9.91	8.71	0.34	-0.65	8.77	11.84	6.73	0.73	5.21	11.17	
60	TRSD					TBYD & TRSD					TRSD & DFSD					TBYD & TRSD & DFSD					
	No Constraint					No Constraint					No Constraint					No Constraint					
	VAR	16.90	37.62	0.27	-1.84	-9.57	18.26	58.81	0.19	-2.93	-187.73	13.66	34.33	0.20	-2.87	-11.63	15.35	35.95	0.24	-2.31	-3.34
	GARCH	15.33	33.69	0.25	-2.07	-10.37	17.93	37.15	0.30	-1.37	-7.07	13.39	37.87	0.17	-3.25	-22.99	11.93	35.19	0.14	-3.69	-9.57
	OPT	11.39	40.83	0.11	-4.18	-39.89	11.65	55.24	0.09	-4.54	-153.5	10.75	36.00	0.11	-4.22	-34.87	10.75	40.86	0.09	-4.42	-18.48
	NLNP	12.39	45.71	0.12	-4.03	-82.39	20.51	32.84	0.42	0.40	3.18	8.59	35.60	0.05	-5.12	-39.26	9.12	36.03	0.06	-4.91	-11.65
	No-Short-Sale					No-Short-Sale					No-Short-Sale					No-Short-Sale					
	VAR	11.46	11.49	0.39	0.14	9.36	12.98	9.85	0.61	3.46	11.51	10.73	11.02	0.35	-0.60	8.78	10.89	6.90	0.57	2.87	10.19
	GARCH	11.73	11.67	0.41	0.40	9.56	12.40	9.84	0.55	2.58	10.92	10.84	10.93	0.36	-0.40	8.92	10.38	8.08	0.43	0.64	9.39
	OPT	10.89	10.89	0.36	-0.32	8.97	11.07	10.90	0.38	-0.08	9.15	10.27	10.98	0.30	-1.22	8.33	10.24	9.53	0.35	-0.55	8.75
NLNP	11.52	11.60	0.40	0.17	9.37	13.58	8.75	0.76	5.66	12.44	11.37	11.89	0.37	-0.18	9.13	9.02	8.49	0.25	-2.07	7.80	

Table 6: Out-Of-Sample Switching Performance

The predictive sample draws are generated from the 5-year expanding estimation. At each month, the portfolio either invests in the market portfolio or 30-day T-bill depending on whether the forecasted expected returns are higher or lower than the riskfree rates. The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns, standard deviations, and two performance measures, Sharpe ratio ($\mathcal{SR}$) and Graham-Harvey measure ($\mathcal{GH2}$), are reported for each predictive variable and model combination. The number of sample draws at each period t is 50000.

Out-Of-Sample Switching Portfolio Performance Tests								
	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$
Benchmarks								
Market	5.82	15.23	0.38					
Fixed	9.63	6.20	0.43	0.75				
Passive	10.18	7.47	0.43	0.74				
Expand	9.96	6.06	0.45	0.95				
DVYD					TRSD			
VAR	8.91	8.89	0.22	-2.45	12.92	12.28	0.49	1.56
GARCH	10.39	11.87	0.29	-1.42	13.71	12.85	0.53	2.16
OPT	7.82	9.31	0.10	-4.37	12.78	12.73	0.46	1.13
NLNP	8.48	9.56	0.16	-3.36	13.28	12.50	0.51	1.85
TBYD					TBYD & TRSD			
VAR	12.66	9.06	0.63	3.72	13.26	11.88	0.53	2.23
GARCH	12.58	9.00	0.63	3.66	13.53	12.51	0.53	2.16
OPT	12.35	8.95	0.60	3.33	12.58	11.95	0.47	1.32
NLNP	11.96	8.90	0.56	2.72	14.39	9.52	0.78	6.01
DFSD					TRSD & DFSD			
VAR	11.22	14.23	0.30	-1.27	11.83	12.26	0.40	0.23
GARCH	13.04	14.79	0.41	0.42	13.54	12.41	0.53	2.23
OPT	10.59	14.16	0.26	-1.91	11.70	12.28	0.39	0.05
NLNP	12.12	14.15	0.36	-0.28	12.90	13.18	0.45	1.02
TBYD & DFSD					TBYD & TRSD & DFSD			
VAR	13.35	8.73	0.74	5.31	10.93	9.88	0.41	0.31
GARCH	12.43	8.62	0.64	3.83	10.95	10.13	0.40	0.19
OPT	11.05	7.34	0.56	2.67	11.87	10.33	0.48	1.42
NLNP	12.86	7.89	0.75	5.55	11.20	10.30	0.41	0.45

Table 7: Out-Of-Sample Predictive Variable Expanding Performance with Transaction Costs

The predictive sample draws are generated from the 5-year expanding estimation. The optimal portfolio weights are calculated from maximizing the expected power utility in the presence of transaction costs. The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns and standard deviations of the realized portfolio returns, and three performance measures, Sharpe ratio ($\mathcal{SR}$), Graham-Harvey measure ($\mathcal{GH2}$), and CER (r_{ce}), are reported for each predictive variable and model combination.

Out-Of-Sample 5-year Rolling Portfolio Performance Tests with Transaction Costs																				
	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$
Benchmarks																				
Market	5.82	15.23	0.38		9.18															
Passive	10.18	7.47	0.43	0.74	9.32															
TB & DEF																				
	No Constraint					$\tau = 0.25\%$					$\tau = 0.50\%$					$\tau = 1.00\%$				
VAR	18.15	41.86	0.27	-1.80	-15.53	17.47	40.35	0.26	-1.90	-12.37	17.27	38.86	0.27	-1.83	-8.95	17.19	37.33	0.28	-1.69	-4.38
GARCH	14.33	29.76	0.25	-2.08	-0.58	13.15	28.57	0.22	-2.55	-0.26	11.64	27.26	0.17	-3.22	-0.22	11.72	23.98	0.20	-2.81	2.99
OPT	9.76	36.62	0.08	-4.66	-13.61	9.40	34.35	0.07	-4.74	-10.60	8.84	32.90	0.06	-4.95	-8.81	5.11	31.58	-0.06	-6.68	-10.10
NLNP	9.05	39.59	0.05	-5.02	-38.06	8.67	35.56	0.05	-5.09	-20.81	7.69	32.00	0.02	-5.46	-12.55	8.29	25.76	0.05	-5.03	-2.20
No-Short-Sale																				
VAR	11.89	7.60	0.65	4.07	11.05	11.85	7.36	0.67	4.31	11.06	11.21	6.62	0.65	3.99	10.57	9.01	7.04	0.30	-1.31	8.25
GARCH	11.52	7.35	0.62	3.63	10.72	11.32	7.17	0.61	3.46	10.56	10.03	6.99	0.44	0.90	9.29	7.69	7.15	0.11	-4.19	6.84
OPT	10.16	6.35	0.51	1.89	9.58	9.88	5.98	0.49	1.66	9.35	9.29	5.80	0.41	0.34	8.79	7.56	6.21	0.10	-4.27	6.93
NLNP	11.84	6.73	0.73	5.21	11.17	11.47	6.26	0.72	5.16	10.89	10.52	5.87	0.61	3.47	10.02	9.18	5.26	0.43	0.70	8.78
TB & TERM																				
	No Constraint					$\tau = 0.25\%$					$\tau = 0.50\%$					$\tau = 1.00\%$				
VAR	18.26	58.81	0.19	-2.93	-187.7	19.63	57.86	0.22	-2.52	-156.9	19.37	55.58	0.22	-2.46	-128.9	18.62	52.02	0.23	-2.44	-80.34
GARCH	17.93	37.15	0.30	-1.37	-7.07	17.63	36.60	0.29	-1.42	-6.20	17.36	35.80	0.29	-1.44	-5.08	18.45	35.62	0.32	-0.96	-3.43
OPT	11.65	55.24	0.09	-4.54	-153.5	4.98	81.43	-0.02	-6.18	-1501	2.61	84.21	-0.05	-6.60	-1766	14.88	51.52	0.15	-3.50	-80.12
NLNP	20.51	32.84	0.42	0.40	3.18	19.79	31.67	0.41	0.29	3.24	20.66	31.95	0.43	0.64	3.75	17.75	30.64	0.35	-0.50	2.03
No-Short-Sale																				
VAR	12.98	9.85	0.61	3.46	11.51	12.31	10.71	0.50	1.78	10.52	12.39	10.88	0.50	1.77	10.56	13.14	11.49	0.54	2.35	11.01
GARCH	12.40	9.84	0.55	2.58	10.92	11.91	11.05	0.45	0.99	9.94	12.18	10.76	0.49	1.56	10.36	12.91	11.07	0.54	2.34	10.91
OPT	11.07	10.90	0.38	-0.08	9.15	11.73	10.97	0.44	0.79	9.78	11.20	11.10	0.38	0.00	9.21	12.59	11.76	0.48	1.46	10.39
NLNP	13.58	8.75	0.76	5.66	12.44	13.68	9.03	0.74	5.47	12.47	13.52	9.16	0.72	5.05	12.27	12.74	10.00	0.58	2.96	11.20

Table 8: Out-Of-Sample Predictive Variable Subperiod Performance

The predictive sample draws are generated from the 5-year expanding estimation. The testing period is divided into four 5-year periods coincident with the re-estimation. Panel A reports the period-by-period performance for each period, whereas Panel B reports the accumulative performance for each period. Performance is reported only for T-bill yield. The average returns, standard deviations, and three performance measures, Sharpe ratio ($\mathcal{SR}$), Graham-Harvey measure ($\mathcal{GH2}$), CER (r_{ce}), and the performance differences between the predictive models and the benchmark (Δ_{gh} and Δ_{cer}), are reported.

Out-Of-Sample 5-Year Subperiod Portfolio Performance Tests															
Panel A: Period-by-Period Performance								Panel B: Accumulative Performance							
$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	Δ_{gh}	Δ_{cer}		$r_p(\%)$	$\sigma_p(\%)$	$\mathcal{SR}$	$\mathcal{GH2}$	$r_{ce}(\%)$	Δ_{gh}	Δ_{cer}	
Expanding Window Benchmark															
1979 - 1983	9.74	6.69	0.05	0.63	10.42			1979 - 1983	9.74	6.69	0.05	0.63	10.42		
1984 - 1988	11.48	7.13	0.25	1.16	7.89			1979 - 1988	10.61	6.89	0.16	0.77	9.15		
1989 - 1993	9.02	4.92	0.43	0.51	7.13			1979 - 1993	10.08	6.29	0.23	0.62	8.47		
1994 - 1998	9.51	5.93	1.10	0.84	10.74			1979 - 1998	9.94	6.19	0.43	0.77	9.04		
TBYD															
VAR								VAR							
1979 - 1983	11.53	3.20	0.25	3.77	11.38	3.14	0.97	1979 - 1983	11.53	3.20	0.25	3.67	11.38	3.05	0.97
1984 - 1988	9.25	7.30	0.33	2.65	14.55	1.49	6.67	1979 - 1988	10.39	5.62	0.29	2.97	9.91	2.20	0.76
1989 - 1993	8.12	6.96	0.39	0.06	6.06	-0.45	-1.07	1979 - 1993	9.63	6.09	0.33	2.18	9.08	1.56	0.60
1994 - 1998	12.61	6.94	1.13	1.32	4.77	0.47	-5.98	1979 - 1998	10.38	6.31	0.55	2.45	9.78	1.68	0.74
GARCH								GARCH							
1979 - 1983	11.04	1.00	0.45	6.85	11.02	6.23	0.61	1979 - 1983	11.04	1.00	0.45	6.56	11.02	5.94	0.61
1984 - 1988	9.93	6.61	0.47	5.34	15.39	4.18	7.50	1979 - 1988	10.48	4.71	0.36	4.25	10.15	3.48	1.00
1989 - 1993	7.71	6.75	0.35	-0.58	6.06	-1.09	-1.07	1979 - 1993	9.56	5.47	0.35	2.56	9.11	1.94	0.64
1994 - 1998	12.19	7.06	1.05	0.08	4.77	-0.76	-5.98	1979 - 1998	10.21	5.90	0.56	2.62	9.69	1.85	0.65
OPT								OPT							
1979 - 1983	11.03	1.00	0.45	6.85	11.02	6.23	0.60	1979 - 1983	11.03	1.00	0.45	6.56	11.02	5.94	0.60
1984 - 1988	9.40	5.38	0.48	5.56	15.09	4.40	7.20	1979 - 1988	10.22	3.86	0.38	4.50	9.99	3.73	0.84
1989 - 1993	7.70	6.69	0.35	-0.57	6.06	-1.08	-1.07	1979 - 1993	9.38	4.98	0.35	2.54	9.01	1.93	0.54
1994 - 1998	12.84	8.86	0.91	-2.04	4.77	-2.88	-5.98	1979 - 1998	10.24	6.17	0.54	2.31	9.66	1.55	0.62
NLNP								NLNP							
1979 - 1983	10.97	0.92	0.45	6.85	10.96	6.23	0.54	1979 - 1983	10.97	0.92	0.45	6.56	10.96	5.94	0.54
1984 - 1988	8.32	3.91	0.39	3.75	14.21	2.59	6.32	1979 - 1988	9.64	2.85	0.32	3.48	9.52	2.71	0.37
1989 - 1993	6.98	6.16	0.26	-1.71	6.06	-2.22	-1.07	1979 - 1993	8.76	4.25	0.27	1.23	8.49	0.61	0.02
1994 - 1998	12.94	8.57	0.95	-1.40	4.77	-2.24	-5.98	1979 - 1998	9.80	5.65	0.51	1.89	9.32	1.13	0.28

Table 9: Out-Of-Sample Mean-Variance Performance

The predictive sample draws are generated from the 5-year expanding estimation. The optimal portfolio weights are calculated from maximizing the expected quadratic utility with a risk aversion coefficient of 4. The realized portfolio returns r_{pt} are calculated from the observed excess returns r_t and the riskfree rates. The average returns, standard deviations, and two performance measures, Sharpe ratio (SR) and Graham-Harvey measure ($\mathcal{GH2}$), are reported for each predictive variable and model combination. The number of sample draws at each period t is 50000.

Out-Of-Sample Mean-Variance Portfolio Performance Tests																
	$r_p(\%)$	$\sigma_p(\%)$	SR	$\mathcal{GH2}$	$r_p(\%)$	$\sigma_p(\%)$	SR	$\mathcal{GH2}$	$r_p(\%)$	$\sigma_p(\%)$	SR	$\mathcal{GH2}$	$r_p(\%)$	$\sigma_p(\%)$	SR	$\mathcal{GH2}$
Benchmarks																
Market	12.74	15.10	0.38													
Fixed	8.65	4.47	0.38	-0.00												
Passive	8.93	5.27	0.38	-0.08												
Expand	8.60	4.36	0.38	-0.02												
DVYD				TBYD				DFSD				TBYD & DFSD				
No Constraint				No Constraint				No Constraint				No Constraint				
VAR	5.29	12.90	-0.13	-7.73	27.10	52.52	0.39	-0.09					28.91	49.22	0.45	0.85
GARCH	7.10	11.82	0.02	-5.59	18.50	39.08	0.30	-1.39	9.67	12.87	0.21	-2.61	19.31	32.60	0.38	-0.13
OPT	5.16	17.30	-0.10	-7.36	21.17	43.08	0.33	-0.88	9.92	14.49	0.21	-2.70	18.03	38.40	0.29	-1.49
NLNP	1.59	17.98	-0.30	-10.30	13.06	24.04	0.26	-1.99	10.18	10.56	0.31	-1.15	18.57	38.42	0.30	-1.28
No-Short-Sale				No-Short-Sale				No-Short-Sale				No-Short-Sale				
VAR	7.48	7.07	0.08	-4.63	9.77	5.78	0.49	1.61					11.55	7.45	0.62	3.57
GARCH	8.35	8.52	0.17	-3.30	9.50	5.23	0.49	1.67	10.26	9.98	0.33	-0.78	11.02	7.02	0.58	3.00
OPT	7.49	7.82	0.07	-4.72	9.62	5.59	0.48	1.50	9.09	10.76	0.20	-2.77	9.99	6.04	0.51	1.88
NLNP	7.96	5.81	0.18	-3.11	9.18	4.88	0.47	1.22	9.36	7.68	0.31	-1.04	11.48	6.49	0.70	4.78
TRSD				TBYD & TRSD				TRSD & DFSD				TBYD & TRSD & DFSD				
No Constraint				No Constraint				No Constraint				No Constraint				
VAR	21.98	37.79	0.40	0.16	36.10	63.48	0.46	1.04	18.13	33.92	0.33	-0.86	22.57	42.84	0.37	-0.36
GARCH	19.36	32.89	0.38	-0.14	25.90	42.20	0.45	0.91	17.51	35.27	0.30	-1.32	19.58	39.55	0.32	-1.03
OPT	17.72	39.02	0.28	-1.67	28.99	59.96	0.37	-0.33	15.25	34.34	0.24	-2.18	21.71	46.42	0.32	-1.06
NLNP	17.91	45.73	0.24	-2.22	22.59	30.06	0.52	2.01	10.19	32.18	0.10	-4.29	16.32	37.62	0.25	-2.09
No-Short-Sale				No-Short-Sale				No-Short-Sale				No-Short-Sale				
VAR	11.12	11.29	0.37	-0.20	12.75	9.37	0.62	3.56	10.45	10.90	0.32	-0.93	10.61	6.54	0.56	2.72
GARCH	11.24	11.47	0.38	-0.13	11.91	9.22	0.54	2.35	10.16	10.74	0.30	-1.27	10.02	7.48	0.41	0.44
OPT	10.21	10.55	0.31	-1.10	10.46	10.69	0.33	-0.83	9.82	10.67	0.27	-1.71	9.79	9.38	0.31	-1.19
NLNP	11.17	11.40	0.37	-0.19	13.11	8.54	0.72	5.12	11.00	11.71	0.35	-0.56	8.42	8.08	0.19	-3.01